

The next leg of AI is about feeding the chips

Does the artificial intelligence (AI) boom feel a little long in the tooth?

ChatGPT will soon blow out the candles on its third birthday. It's been more than two years since **Nvidia's (NVDA)** blockbuster earnings lit the fuse on what's become the biggest disruptive megatrend of our lives.

I (Stephen) understand the instinct to start looking for "the next thing." But a decade of investing in disruption has taught me the hardest lesson in markets: megatrends last longer than you think, and the money is made by letting winners run.

AI is still *the* investing megatrend of our time.

From using AI tools every day and watching the dollars flow, it's clear to me that this buildout is only getting started.

When the music's turned up to volume 10, you don't sit out. You dance.

We've already made a lot of money being early to the AI trade. But the deeper I dig, the clearer it becomes we're still in the early innings of a multi-decade cycle.

But the money is shifting, and that's where the next wave of opportunity lies.

In this month's issue

Are we in a bubble?

The first AI wave was "buy GPUs..."

Memory: from commodity to crown jewel

HBM solves the "memory wall" problem

One supplier, many tailwinds

Key prices

S&P 500	6,666.83
1-Month (Gain) +3.92%	
Nasdaq	22,568.64
1-Month (Gain) +6.06%	
Russell 2000	2,435.67
1-Month (Gain) +3.55%	
Dollar Index	97.51
1-Month (Loss) -0.90%	
US 10-Year Yield	4.092%



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Are we in a bubble?

Longtime *Disruption Investor* members know the AI buildout is the largest infrastructure buildout of our lifetimes. This year alone, big tech will spend north of \$350 billion building AI data centers.

Many investors think this is a bubble ready to pop at any moment. Some common pushbacks I hear, and my take:

"Where's the revenue?"

The biggest AI spenders are currently spending way more on AI than they're earning. But before revenues comes usage, which is surging.

ChatGPT is the fastest-growing product in human history. It now has over 700 million weekly users. **Google's (GOOGL)** Gemini recently rocketed to the top of the App Store.

"They're overspending."

Listen to what the CEOs of the largest and most profitable companies on Earth are saying. Google's Larry Page has said internally many times that he's willing to go bankrupt rather than lose this race.

And **Meta Platforms (META)** founder Mark Zuckerberg said:

If we end up mispending a couple hundred billion dollars, I think that is going to be very unfortunate, obviously. But what I'd say is I actually think the risk is higher on the other side.

Anyone trying to call the top of the AI spending boom needs to remember those two quotes.

The people spending the money aren't optimizing near-term "return on investment." They believe the prize is a "digital-god-scale" platform worth trillions over decades. When the prize is that large, the rational move is to keep investing.

Don't you want to be on the other end of this firehose of money? I do!

Owning the disrupters getting paid *now* has been our strategy from Day 1, and it's worked beautifully. We'll keep running that playbook.

"They'll run out of cash."

These companies have so much spare cash lying around, they don't know what to do with it.

But the opportunity is *shifting*. We're entering Phase 2 of the boom, and it's minting a new class of winners.

The first AI wave was “buy GPUs...”

The next wave is “feed them.”

Data centers aren't just endless rows of Nvidia GPUs. They're three projects running at once: the brains, the plumbing, and the real estate.

For every \$100 spent on a new AI data center, here's roughly where the money goes:

The brains: \$55–\$65. AI chips, plus the high-speed networking gear, fiber optics, and switches that let chips talk to each other at light speed.

Power and cooling: \$25–\$35. All the equipment that gets electricity in and heat out. Giant transformers and the new generation of liquid-cooling loops that keep racks from melting down.

Land and buildings: \$10–\$20. The dirt, the concrete, and the pipes. Buying the site and putting up the shell of the building.

Three years ago, roughly 75% of AI spend went toward GPUs. Today, that's closer to 50% and falling. That's because we're shifting from training (teaching the AI model) to inference (using the AI model).

When you're spending \$350 billion/year on data centers, that subtle change is massive. And one of the biggest winners from this shift are memory chips.

There are two main types of computer chips:

#1: Logic: Nvidia GPUs, which process data.

#2: Memory, which stores data.

Running AI models requires logic chips to process oodles of data, but also memory chips that can store and release it quickly. That's why AI servers are memory hogs.

A typical AI server chews through 8X more memory than a classic computer. Each generation keeps ratcheting up. Nvidia's upcoming Rubin accelerator is designed with roughly 4X more memory than its predecessor.

On Nvidia's latest Blackwell chips, about 60% of the total manufacturing cost is memory. The “brains” aren't the only story anymore. The memory bill has become bigger than the compute bill.

Imagine an AI system as a kitchen...

The GPU is the chef; memory is the pantry.

If the pantry is far away, the chef wastes time jogging back and forth to grab ingredients. But if it's right beside the stove, the chef can cook much faster.

The big bottleneck in AI right now is that chips are fast but feeding them data isn't. More than 90% of the time it takes an AI model to respond is spent shuttling data back and forth between logic and memory chips, according to SK Hynix.

The industry calls this the "memory wall." Compute keeps getting faster, but memory hasn't kept up. So \$50,000 chips sit idle waiting for data.

As models get larger and try to "remember" longer passages of text, the memory they need to keep on hand balloons even more.

To make better AI, we need a *new* way to feed them. Enter High-Bandwidth Memory (HBM).

HBM is a fancy term for stacking memory chips like a skyscraper and bolting them right next to the GPU. This means data only has to travel millimeters, not inches.

Shorter distance = more data per second = better AI. That's why every next-gen AI system is being designed around HBM.

For almost two decades, memory was the dog of the chip world. Now, it's fast becoming the new kingmaker of Phase 2 of the AI buildout.

Three shifts are driving a new memory supercycle

#1: Conversations got long. Models that once handled a short paragraph now chew through entire documents. Long context means the live memory a chip needs on hand balloons. The longer the exchange, the more memory you need right beside the GPU. That's what HBM delivers.

#2: Usage exploded. Google now processes 980 trillion AI "tokens" every month. AI agents are booking travel and carrying out science experiments. Each task consumes multiple times more tokens than a simple chatbot. More tokens equal a need for continuous, always-on memory. That pushes relentless demand for HBM.

#3: New buyers. Nvidia remains the #1 customer for HBM. But the custom chips from Google, **Amazon (AMZN)**, and **Broadcom (AVGO)** are rapidly snapping up supply.

Combined, these three shifts tell us we're entering a new memory supercycle.

The lesson in megatrends is to own the bottleneck. Phase 1's bottleneck was GPUs. Phase 2's bottleneck is HBM.

Most investors are still staring at Nvidia (we've kept a slice of NVDA to capture that upside).

But the real story is a "forgotten" memory maker that's about to go from zero to hero, just like Nvidia did when it pivoted from a gaming card company into the backbone of the AI boom.

Here's Chris with the full rundown...

Memory: from commodity to crown jewel

Not long ago, the computing world viewed memory chips as commodities.

They were interchangeable parts, made by multiple companies. Buyers treated them almost like bulk wheat or oil. Prices rose and fell quickly, with swings in supply and demand.

It didn't matter who made the DRAM or NAND chips used in smartphones, personal computers, and data centers, as long as they were compatible.

DRAM—short for dynamic random-access memory—are super-fast memory chips that temporarily hold data (programs and files) in active use for processors (CPUs, GPUs, etc.) to grab quickly. Think of it like a digital scratchpad—great for temporary notes, but it's “volatile,” so all that data vanishes when the power's turned off.

NAND chips are a type of flash memory that store data even when the power's off, like a digital filing cabinet.

Data stays put until you rewrite or erase it, making NAND perfect for long-term storage in devices like smartphones, USB drives, and the solid-state drives (SSDs) that have replaced—and are much faster than—the traditional hard disk drives of the past.

For example, a server manufacturer like **Dell Technologies (DELL)** could buy standard DRAM chips or modules—bigger plug-and-play packages that house multiple chips on a circuit board—from Samsung, SK Hynix, **Micron Technology (MU)**, or others, as long as they met the desired specs and price.

This led to boom-bust cycles:

- Periods of high demand drove memory prices up.
- Memory manufacturers overproduced to meet that demand.
- And the resulting memory glut sent prices crashing down.

Investors in memory companies—like us—understood this dynamic. So we typically wouldn't pay a high multiple of earnings (or revenue, cash flow, or EBITDA) for memory stocks, because of the industry's cyclical nature. *Keep this in mind for later.*

But the demands of AI have completely changed the commodity-like nature of memory. It's no longer an afterthought or interchangeable part of computing. It's a high-tech, customized core component that's crucial to consider in the earliest stages of AI system design.

Modern AI systems need memory that's tightly integrated with processors. So much so that memory now accounts for a majority of AI chip costs in many cases.

As we mentioned earlier, memory makes up 60% of Nvidia's new Blackwell AI chips' total manufacturing cost. In other words, the "pantry" now costs more than the "chef."

This is a radical shift. The brains are no longer the only star of the show. Memory has become just as critical.

This shift happened because feeding data to AI systems faster—and in larger volumes—yields huge improvements in capabilities.

AI models are hungry for data, and lots of it, immediately. If memory can't keep up, even a \$50,000 state-of-the-art AI processor sits idle waiting for data.

Think about ChatGPT and its 700 million weekly users.

Each query from each of those 700 million users involves the AI model juggling and storing a lot of data in memory to generate a response.

Meanwhile, new AI applications and agents—from assistants that book travel for you to systems that analyze scientific data on a scale that humans can only dream of—are doing much more complex, multi-step tasks that require keeping even more data readily accessible in memory.

This skyrocketing demand has created a bottleneck opportunity in memory.

HBM solves the “memory wall” problem

As we said earlier, “The lesson in megatrends is to own the bottleneck.” We did that with Nvidia when GPUs were the bottleneck.

Now, we're going to do it with **Micron Technology (MU)** when HBM is the bottleneck.

Micron is one of the few companies in the world capable of producing HBM.

In older AI chips designs, multiple DRAM memory chips were positioned around the central processor on the circuit board. Data had to travel along wires on the board between the GPU and these external memory chips.

In new AI chips, this paradigm has changed—memory is now stacked right next to or on top of the processor in the same package, eliminating the long travel distance.

The result of this integration is HBM. It takes several DRAM memory chips and stacks them vertically—essentially forming a 3D memory cube—and connects them directly to the processor with tiny wires called through-silicon vias rather than through a traditional circuit board path.

It's like replacing our separate pantry room with a pantry cabinet attached to the stove. The distance for data to travel is vastly reduced and the width of the "pipe" between the processor and memory is vastly increased. This means orders of magnitude more data can be fed to the processor per second.

And HBM isn't just faster. It's also more efficient (less energy wasted per bit of data transferred) because the shorter distance and wide bus mean it can run at lower clock speeds for the same throughput.

Put simply, HBM is the answer to the "memory wall" problem we talked about earlier.

Every new high-performance AI chip now comes with HBM built in. These chips can't function without it. You can't swap in some generic memory later. The memory is part of the product from the start.

This tight coupling flips the script on how memory is procured and priced.

Instead of shopping for generic DRAM modules on the open market, companies developing AI chips—like Nvidia, Google, Amazon, and **Advanced Micro Devices (AMD)**—must work hand-in-hand with memory suppliers during the design phase to ensure they get the right HBM. And they must enter into supply agreements well in advance to secure enough of it.

For example, Micron already has pricing agreements in place with six big customers for most of the HBM it will be able to make in 2026.

It's worth noting that one of those customers is Nvidia, and that **Micron's HBM is integrated into Nvidia's Blackwell Ultra B200 GPUs (cutting-edge AI chips), which power the AI clusters of hyperscalers like Google, Amazon, and Microsoft (MSFT).**

We expect Micron to ink agreements for its remaining 2026 HBM supply before the end of 2025.

In other words, by the end of this year, Micron should have already pre-sold all the HBM it's capable of producing next year at agreed prices.

Traditional DRAM never had that luxury. Prices used to swing quarter to quarter. But with HBM, long-term contracts and sold-out supply are becoming the norm, keeping prices much higher and much steadier.

The new "memory supercycle" we mentioned earlier refers to a prolonged period of rising demand and constrained supply, allowing memory makers to enjoy strong pricing and profits.

Unlike past cycles—driven by things like PC upgrades and smartphone adoption—that fizzled, this new memory supercycle is underpinned by AI, a paradigm shift in computing that we believe is multi-decade in scope. As long as AI capabilities keep improving, there will be an insatiable appetite for more memory.

Micron finds itself at ground zero of this tectonic shift.

As one of only three major DRAM manufacturers on the planet—and one of only two companies including SK Hynix currently shipping cutting-edge HBM in volume—Micron is positioned to reap huge rewards from memory's new starring role in AI.

It's worth highlighting that HBM commands a premium price and gross profit margin relative to standard memory.

Because it's so much more complex and advanced—and is now an invaluable part of the AI processor solution—Micron can charge significantly more per gigabyte than for commodity DRAM. Estimates suggest HBM carries about 5X to 6X higher dollar content per bit than normal DRAM.

Meanwhile, Micron's management has indicated that gross profit margins on HBM are well above the corporate average. In Micron's fiscal 2025, which ended August 28, 2025, the company's overall gross profit margin was 39.8%.

However, its data center business—which includes HBM and reached a record 56% of total company revenue in fiscal 2025—had a gross profit margin of 52%.

So as HBM becomes a bigger portion of Micron's sales, it lifts the profitability of the whole company. This is a key reason why Micron's earnings are now growing so fast and will continue to do so.

It's not just about higher sales volume. It's also about higher profit per sale from a richer product mix, thanks to HBM.

According to Micron CEO Sanjay Mehrotra, "HBM revenue grew to nearly \$2 billion" in the most recent quarter and played a big part in the 46% year-over-year growth we saw on Micron's top line.

But the company generated more than \$11.3 billion in total revenue for the quarter. So more than \$9 billion came from sources other than HBM.

One supplier, many tailwinds

As we alluded to earlier, Micron makes two main types of memory chips:

#1: The super-fast DRAM chips that temporarily hold data in active use for processors.

#2: The flash memory NAND chips that store data when the power's off and serve as the building blocks of SSDs.

DRAM revenue, which includes HBM, grew 69% year over year last quarter to a record \$9 billion and represented 79% of total revenue.

For the entire fiscal-year 2025, DRAM revenue grew 62% from the prior year to a record \$28.6 billion. NAND revenue grew 18% to a record \$8.5 billion over the same period.

In addition to data centers, customers use these chips in PCs, smartphones, and automobiles—everywhere memory is used. This diversification is important. It means Micron can ride multiple waves of demand. And right now, many of those waves are rising together, largely thanks to AI.

Cloud companies like Google, Microsoft, and Amazon use enormous amounts of Micron's DRAM and HBM, as well as the company's high-performance SSD storage (made using its NAND chips) in their AI data centers.

AI is having a big impact on smartphone memory, too. The newest generation of phones are touting AI features for things like photography, voice assistants, and on-device chatbots. And these features need more memory.

High-end smartphones now ship with 12GB to 16GB of DRAM, up from 8GB or so just a year or two ago. And some flagship, next-generation AI phones are testing 24GB of DRAM. That's 3X the DRAM of high-end smartphones just a couple years ago.

The phones now need so much memory because when they're running local (on-device) AI models to bypass the cloud for privacy and/or speed, they need a lot of fast memory to hold the model data.

Micron, as a top supplier of mobile DRAM, will benefit as smartphones double or triple their memory capacity in the near term.

Meanwhile, after years of stagnation, the humble PC is also getting a boost from AI. Microsoft's newest Windows releases are integrating AI assistants and features that can summarize documents, edit media, and do other cool things without needing to access the cloud.

These AI-enabled PCs need more memory (DRAM) and more storage (SSDs built from NAND chips) to work smoothly. It's estimated that new PCs with AI capabilities need about 60% more DRAM on average than a standard PC.

Then there's the growing opportunity in automobiles.

Modern vehicles, especially electric vehicles and those with advanced driver-assistance systems (ADAS) or self-driving features, are essentially computers on wheels. They run AI models for vision (to identify lanes, pedestrians, signs, etc.), voice control, smart navigation, and more.

All this requires memory. A car with ADAS needs 2X to 3X more DRAM and NAND flash storage than one without it. That's because it needs to access and make sense of data from camera feeds, radar, and other sensors—and run neural networks in real time without accessing the cloud.

As a leading supplier of automotive memory, Micron stands to benefit handsomely from the explosion in memory content per vehicle as fully autonomous driving matures.

Now, as we alluded to earlier, Micron is one of the “big three” companies in the DRAM and HBM space. The other two are Samsung and SK Hynix. Combined, they control more than 95% of the HBM market.

You might be wondering why we don't just combine all three companies into a makeshift “ETF,” like we did for cybersecurity in [last month's AI issue](#).

There are a number of reasons for this. Here are some of them:

- Samsung is the world's largest memory chip producer, with enormous scale and resources, but it's not a memory “pure play” like the other two. It also makes logic chips, displays, smartphones, TVs, appliances, and more. So the memory boom won't be as big of a needle-mover from Samsung as it will be for Micron and SK Hynix.
- SK Hynix is the world's second-largest memory maker, but it's tough for American investors to buy the stock. Like Samsung, its primary listing is on the Korea Exchange, and it does not have a liquid way for US investors to purchase shares over-the-counter (OTC). SK Hynix's US OTC ticker is supposedly HXSCL, but not a single share of it has traded this year according to my Schwab account.
- Micron is gaining ground and has now surpassed Samsung in global HBM shipments. In the most recent quarter, Micron overtook Samsung in global HBM shipments, capturing a 21% market share compared to Samsung's 17%.
- Micron and SK Hynix are the only two companies currently shipping the most cutting-edge HBM in volume. Samsung is behind here because it took the company longer to pass the necessary qualification tests for this product.

The final thing we want to hit on:

Earlier, we said “keep this in mind for later” when we talked about memory’s historical boom-bust cyclical nature. And that investors like us typically wouldn’t pay a high multiple (of earnings, revenue, cash flow, or EBITDA) for memory stocks.

However, in this new memory supercycle, which simply wasn’t possible before AI and HBM and could easily last five years or more (with some bumps along the road, of course)—memory is going to be more of a steady, fast-growth business than a cyclical business.

This will likely result in investors being willing to pay higher multiples for these stocks, resulting in even higher upside for Micron.

Meanwhile, in the past year, Micron generated \$37.4 billion in revenue, which reflected annual growth of 48.9%. EBITDA, net income, and Real Cash Flow all grew north of 100% in the same period, ringing in at \$18.2 billion, \$8.5 billion, and \$499 million, respectively.

As one of only three companies capable of producing cutting-edge HBM and the sole US-based memory manufacturer, Micron has emerged as a big beneficiary of the memory market’s AI-driven transformation.

What was once a commoditized, low-margin input is now a crucial strategic component in AI systems, commanding premium prices, high margins, and multi-year supply agreements.

With its fiscal 2025 revenue surging nearly 50% and HBM revenue approaching an \$8 billion annual run rate, Micron is successfully capitalizing on the “memory supercycle” driven by insatiable AI demand across data centers, smartphones, PCs, and automotive applications.

Action to take: Buy Micron Technology (MU) at current market prices and set a 60% hard stop.

One of our holdings will report earnings before the next *Disruption Investor* publishes on November 6:

- **Taiwan Semiconductor (TSM):** October 16

Please note that the date listed above is subject to change.

And for the most up-to-date guidance on our current positions, [click here](#) to view the portfolio online.



Chris Wood and Stephen McBride

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